

End-Sem Exam, Probability & Statistics (MCC505)
 Monsoon Semester: 23-24,
 Time: 8:00 am - 11:00 am, 22nd November, 2023,
 Full Marks: 48
 Dept: Mathematics and Computing, IIT (ISM) Dhanbad
 Note: Show all steps in your solution

1. (a) The sum and sum of squares corresponding to length X (in cms.) and weight Y (in gms.) of 50 tapioca tubers are given below:

$$\sum_{i=1}^{50} X_i = 212, \sum_{i=1}^{50} X_i^2 = 902.8, \sum_{i=1}^{50} Y_i = 261, \sum_{i=1}^{50} Y_i^2 = 1457.6$$

Which is more consistent, the length or weight?

mark: 2

- (b) The first moment about origin of a distribution is 5. The second and third moments about the mean are 20 and 140 respectively. Find the third moment of the distribution about 10.

mark: 2

2. (a) Show that for a frequency distribution, $\beta_2 \geq 1$.

mark: 3

- (b) For a frequency distribution, the various calculations are given below: $U = (X - 6)$, $\sum f = 300$, $\sum fU = -61$, $\sum fU^2 = 1445$, $\sum fU^3 = -1285$, $\sum fU^4 = 16469$

For variable X , calculate first four moments about mean, β_1 and β_2 and comment on the nature of skewness and kurtosis of the distribution.

mark: 5

3. (a) Find the moment generating function (mgf) for the distribution with pdf

$$f(x) = \begin{cases} \frac{3}{k^2}, & 0 < x < k \\ 0 & \text{otherwise.} \end{cases}$$

Then, obtain its variance.

mark: 4

- (b) The joint density of X and Y is

$$f(x, y) = \begin{cases} 6x, & 0 < x < 2, 0 < y < (1 - x) \\ 0 & \text{otherwise.} \end{cases}$$

- i) Show that X and Y are not independent.

- ii) Find $P(X > 0.3 | Y = 0.5)$.

mark: 4

- (c) A fair coin is tossed 1000 times. Use the Central Limit Theorem to approximate the probability of getting heads between 480 and 520 times. How does this compare to Chebyshev's bound?

mark: 4

$$P(0 < Z < 1.27) = 0.3980$$

4. (a) Let X_1, X_2, X_3 be a simple random sample from Exponential distribution with mean $\frac{1}{2}$. Derive the joint cdf and pdf of $X_{(1)}, X_{(3)}$. (Show all steps)

mark: 4

- b). Let X_1, X_2, X_3 be a simple random sample from a discrete distribution with pmf $P(X = x) = \frac{1}{10}$ for $x = 1, 2, \dots, 10$. Find $P(X_{(3)} = 8)$.

mark: 3

5. In an industrial process, the diameter of a ball bearing is an important measurement. The buyer sets specifications for the diameter to be 3.0 ± 0.01 cm. The implication is that no part falling outside these specifications will be accepted. It is known that in the process the diameter of a ball bearing has a normal distribution with mean 3.0 and standard deviation 0.005. Find out the probability that out of 10 ball bearings atmost one will be rejected.

[Given $P(0 < Z < 2.4) = 0.4937$, $P(Z < -2.0) = 0.0228$, $P(0 < Z < 1.4) = 0.4192$ where $Z \sim N(0, 1)$]

mark: 5

6. (a) Let X and Y be independent random variables with standard deviations σ_x and σ_y . Show that the correlation coefficient between X and $(X+Y)$ is $\frac{\sigma_x}{\sqrt{\sigma_x^2 + \sigma_y^2}}$.

mark: 5

- (b) Prove that regression coefficients are invariant under origin change but affected by scale change.

mark: 2

- (c) In a partially destroyed laboratory record of an analysis of correlation data, the following results only are legible:

$\sigma_x^2 = 9$, Regression line of Y on X : $8x - 10y + 66 = 0$ and Regression line of X on Y : $40x - 18y = 214$. Find the mean of X and Y , r_{xy} , and σ_y^2 .

mark: 5